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Exercise 1: SQL Fundamentals - SELECT & Filtering.

Course : BrightLearn Basic SQL

Questions : 10 queries on 1 table

Focus : SELECT • WHERE • ORDER BY

01. Selecting and Sorting

01 Retrieve all columns from the employee table.

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary,  
       hire_date,  
       city  
FROM employees;
```

Results
employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

02 Find all unique departments

```
SELECT DISTINCT department
FROM employees;
```

Results

Department
IT
HR
Finance
Marketing

03 Retrieve first and last names ordered by salary descending

```
SELECT first_name,
       last_name
FROM employees
ORDER BY salary DESC;
```

first_name	Last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04 Retrieve the top 3 highest-paid employees.

```
SELECT id,  
       first-name,  
       last-name,  
       salary.  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

Results

id	first-name	last-name	salary
5	Eva	Naldoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02 Filtering with WHERE

05 Find employees in the IT department

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM employees  
WHERE department = 'IT';
```

Results

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naldoo	IT

06 Find employees in Finance with salary > 60000.

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary  
FROM   employees  
WHERE  department = 'Finance' AND salary > 60000;
```

Results

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

07 Find employees in HR or Marketing

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM   employees  
WHERE  department = 'HR' OR 'Marketing';
```

Results

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokaend	Marketing

08 Find employees not in IT

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE NOT department = 'IT';
```

Results

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09 Find employees in IT, HR or Finance using IN

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department IN ('IT', 'HR', 'Finance')
```

Results

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10 Find employees in IT with salary > 65000 and city Johannesburg

```
SELECT *  
FROM employees  
WHERE department = 'IT' AND salary > 65000 AND city = 'Jhb';
```

Results

id	first_name	last_name	department	salary	Hire date date	City
1	Alice	Green	IT	70000	2020-01-10	JHB
5	Eva	Naldo	IT	72000	2017-09-30	JHB